The Proton as a Vibrating Torus: Geometric Foundations of Mass Emergence in FFGFT

Contents

Summary

In FFGFT the proton is not described as a collection of quarks and gluons in a volume, but as a *vibrating torus* $T^4/\mathbb{Z}_3$. Quarks and gluons are different mode types of this structure. The proton mass is the vibrational energy of the torus. This document develops the geometric picture, anchors it in the corpus, and marks all statements by the epistemic standard [K]/[S]/[SETZUNG].

Reading note: inner perspective

[K] We describe the proton torus *from inside* – as part of the same T^4 system (Doc. 248). There is no outside perspective. What "confinement" or "freedom" means is relational: it describes the relation between the observer and the respective topological system, not an intrinsic property of the observed field.

.1 The Geometric Picture

[K] FFGFT starts from a compact 4D torus T^4 with $\mathbb{Z}_3$ orbifold symmetry. The fundamental relation is (Doc. 003):

$$\tilde{T} \cdot m = 1 \tag{1}$$

Intrinsic time and mass are dual. This has immediate consequences for all modes on the torus.

[K] The triality of the D4 lattice is built into the lattice geometry, not postulated (Doc. 314): the order-3 elements in Aut$(D4)$ are *half-integer* – the glue-vector mixture $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ enters. The $T^4/\mathbb{Z}_3$ structure with exactly 9 fixed points follows from the D4 geometry. These half-integer elements are the geometric origin of the fermionic statistics of the quark modes.

.2 Two Mode Types: Quarks and Gluons

[K] The compact $\mathbb{Z}_3$ fibre carries the discrete, mass-bearing spectrum – not the unrolled axis (Doc. 285):

> "The discrete, mass-bearing spectrum sits on the compact $\mathbb{Z}_3$ fibre, not on the unrolled axis."

Table 1: Quark and gluon modes on the torus [K]

Mode	Spectrum	Mass	Proper time $\tilde{T}$	Doc.
Quark modes	discrete, $\mathbb{Z}_3$ fibre	$m > 0$	$\tilde{T} < \infty$	006, 285, 314
Gluon modes	continuous, locally flat	$m = 0$	$\tilde{T} = \infty$	003, 270

[K] Gluons are massless ($m = 0$), hence $\tilde{T} = \infty$ – they perform no measurement act (Doc. 270). Locally, flatness holds: "in the spatial limit, local flatness" (Doc. 270). This means: locally the massless gluon sees $\mathbb{R}^3$, but that is a local homeomorphism, not a global statement.

.3 The Relational Nature of Confinement and Freedom

[K] Unrolling is a measurement act of the embedded observer (Doc. 270): "The measurement chooses one circle as time and unrolls it; a physical act, not a mathematical limit."

[K] We (observers) are part of the T^4 system – not outside (Doc. 248): "We are part of the system. We always describe from inside, never from outside."

This yields a fundamental insight for the photon-gluon distinction:

Confinement is relational [K/S

] Photon and gluon are both massless ($m = 0$), both timeless ($\tilde{T} = \infty$), both see $\mathbb{R}^3$ locally. **Both carry boundary conditions.**
The difference is not intrinsic but *relational*:

- **Photon:** We (observers) are part of the same T^4 system. The photon inhabits the same topological sector. Its boundary condition is our boundary condition. It therefore appears to us as freely propagating.
- **Gluon:** We face the $\mathbb{Z}_3$ substructure of the proton torus from outside. The gluon inhabits a different topological sector. It appears to us as confined.

"Confinement" is not a property of the gluon alone – it is a property of the relation observer ↔ system (Doc. 248, [K]). **[S]** The complete topological derivation of why photon and gluon inhabit different sectors remains open.

.4 Massless Field Modes: Locally Pure Energy

[K] For massless fields ($m = 0$), $\tilde{T} \cdot m = 1$ is not defined – not as a limit, but exactly: on null geodesics $d\tau = 0$; along a light path no proper time elapses. There is no rest frame (Doc. 312):

"Masslessness means $v = c$ exactly – the saturation of the inequality $v \leq c$. There is no $c^*(\lambda)$; $dc/d\lambda \equiv 0$."

[K] What remains locally is exclusively $E = pc$ – pure energy (Docs. 134, 312). This holds for photon field modes and gluon field modes *equally*. From a local point of view, every massless field mode is pure energy. There is no "gluon here" or "photon there" in an absolute sense – only energy density in space.

[K] "Particles" are not fundamental in FFGFT (Docs. 049, 156): "There are no particles at all! What we call particles are different excitation patterns in the single field $\delta m(x,t)$." Photon and gluon are field modes – not objects.

[S] The difference photon/gluon lies not in the local description (both: $E = pc$, both: $d\tau = 0$, both: locally flat), but in the *global topological embedding* of the field mode:

Field mode	Topology	Global behaviour
Photon	$U(1)$ flux, extended	propagates in T^4 (our system)
Gluon	$\mathbb{Z}_3/SU(3)$ linking	standing wave in proton torus

[S] The gluon field is not localisable: as a non-Abelian $SU(3)$ field mode (Doc. 145: colour-anticolour linking of three flux threads) it interacts with itself. There is no bounded "gluon object" – only a global field configuration of the proton torus. The eight gluon modes correspond to the eight non-trivial linking states of the three flux threads = eight generators of $SU(3)_c$ (Doc. 145, [S]).

.5 Topological Phase Closure as Confinement Mechanism

[K] On the closed torus a wave must complete an integer phase shift over the circumference:

$$\oint k\,\mathrm{d}l = 2\pi n, \quad n \in \mathbb{Z} \tag{2}$$

[K] The $\mathbb{Z}_3$ action on $L^2(T^4)$ permutes the mode orbits cyclically; the permutation matrix has eigenvalues $\{1, \omega, \omega^2\}$, phases 0°, $+120^\circ$, -120° – structurally exactly the circulant diagonalisation of the $\mathbb{C}^3$ fibre (Doc. 314). This enforces discrete modes without walls.

[S] The step from this mode structure to full QCD confinement dynamics has not yet been formally carried out.

.6 The Duality $\tilde{T} \cdot m = 1$ in the Proton

[K] The complete energy relation $E^2 = (m_0c^2)^2 + (pc)^2$ (Doc. 318) shows: massless gluons carry energy $E = pc$ despite $m_0 = 0$ – they carry $\sim 99\,\%$ of the proton mass as field energy.

[K] This is not a paradox. A standing wave has no proper time ($\tilde{T} = \infty$), but its energy $E = \hbar\omega$ is real and measurable. The duality $\tilde{T} \cdot m = 1$ states: timelessness and masslessness are equivalent. The *system energy* of the bound state – not the eigen-energy of individual modes – is the proton mass.

.7 Spin Statistics from D4 Triality

[K] The fermionic statistics of the quark modes follows from the D4 lattice geometry (Doc. 314): the 80 order-3 elements in Aut($D4$) fall into three classes; the class with trace -2

Table 2: Time-bearing vs. timeless oscillation [K]

Property	Quark (time-bearing)	Gluon (timeless)
Proper time	$\tilde{T} > 0$	$\tilde{T} = \infty$
Energy	Rest mass + motion	Pure field energy $E = pc$
Spectrum	Discrete ($\mathbb{Z}_3$ fibre)	Locally continuous
Statistics	Fermion (D4 triality [K])	Boson

is *half-integer* and fixed-point-free in $\mathbb{R}^4$. This half-integrality is the geometric foundation of fermion statistics.

[S] The draft approach "$(n_\theta + n_\phi)$ odd → fermion" captures the result but not the path. Correct is: D4 triality delivers half-integer representations, independently of the parity of the winding numbers.

.8 Colour Charge and $\mathbb{Z}_3$ Structure

[K] The $\mathbb{Z}_3$ triality of the D4 lattice is built in, not postulated (Doc. 314). $\mathbb{Z}_3$ is the centre of $SU(3)$.

[S] Whether and how the full $SU(3)_c$ gauge structure emerges from this has not yet been formally derived in the corpus.

Status of the colour charge thesis [S

] The statement "colour charge is effective, not fundamental" is a geometric picture **[S]**, not a proved derivation. The $\mathbb{Z}_3$ connection [K] is a structural hint at the $SU(3)_c$ structure – the complete geometric derivation remains open.

.9 Λ_{QCD} and Minimal Frequency

[S] The minimal frequency on a torus with radius $R \approx 1$ fm:

$$\omega_{\text{min}} = \frac{1}{R} \approx 197\ \text{MeV} \approx \Lambda_{\text{QCD}} \tag{3}$$

($1\ \text{fm}^{-1} = 197.3$ MeV in natural units). Consistent with Doc. 041 ($\Lambda_{\text{QCD}} = v \cdot \xi^{1/3} \approx 200$ MeV), but not a complete derivation.

.10 The Proton as a Whole

[K] The proton is not a container with particles but a *system quantity*: the quark modes (discrete, $\mathbb{Z}_3$ fibre, massive) define the boundary condition. The gluon modes (locally continuous, massless) fill the system with field energy. The proton mass is the total energy of this coupled system.

[S] The geometric picture "quarks at the boundary, gluons in the volume" is conceptually robust but not yet a formal derivation from the torus geometry.

.11 Relation to the Corpus

- **Doc. 003**: $\tilde{T} \cdot m = 1$, $D_{\text{frak}} = 2.94$, K_{frak} [K]
- **Doc. 248**: Embedded observer, inner perspective [K]
- **Doc. 270**: Measurement as unrolling, local flatness [K]
- **Doc. 285**: Discrete spectrum on $\mathbb{Z}_3$ fibre [K]
- **Doc. 311**: Rolled-up/unrolled, three ways [K]
- **Doc. 314**: D4 triality, half-integer, $T^4/\mathbb{Z}_3$ from D4 [K]
- **Doc. 318**: Proton as open bridge, $E^2 = (m_0c^2)^2 + (pc)^2$, scope [K]
- **Doc. 041**: $\Lambda_{\text{QCD}} \approx 200$ MeV [S]
- **Doc. 049**: Confinement formulation [S]

.12 Open Questions [S]

1. $SU(3)_c$ **emergence from** $\mathbb{Z}_3$ **triality.** The structural hint [K] is established. The full gauge structure derivation is missing.
2. **Why photon and gluon inhabit different sectors.** The relational explanation (Doc. 248) is [K]. The formal topological derivation remains [S].
3. Λ_{QCD} **from torus geometry.** Doc. 041: order of magnitude [S]. Complete derivation open.
4. **RGE bridge** $\alpha_s(m_\tau) \to \alpha_s(M_Z)$. Doc. 160: $\alpha_s(m_\tau) = 3\xi^{1/4}$ [K]. Bridge open [S].
5. **Formal boundary conditions on** $T^4/\mathbb{Z}_3$. Exact spectral theory still to be developed [S].